Data quality profile - PD / portfolio data

The MEANS Procedure

Variable	N	N Miss	Mean	Minimum	Maximum
age	100	0	42.5200000	21.0000000	70.0000000
employment_years	96	4	7.1531250	0.5000000	20.2000000
annual_income	97	3	180597.94	180000.00	228000.00
credit_limit	100	0	326285.00	50000.00	1332000.00
outstanding_balance	100	0	228256.00	9600.00	1332000.00
monthly_debt_payment	100	0	4260.00	800.0000000	9400.00
current_dpd	100	0	11.4000000	0	74.0000000
default_flag	100	0	0.2200000	0	1.0000000

Data quality profile - PD / portfolio data

n_pd
100

Data quality profile - PD / portfolio data

n_lgd
100

Data quality profile - PD / portfolio data

n_ead
100

Engineered features

The MEANS Procedure

Variable	N	Mean	Std Dev	Minimum	Maximum
utilization	100	0.6340526	0.3177452	0.0568562	1.0000000
dti	100	0.2830211	0.1164199	0.0533333	0.6266667
account_age_months	100	48.2800000	25.0260793	4.0000000	97.0000000
total_delinq_12m	100	1.1100000	1.5498126	0	7.0000000
delinq_severity	100	1.5000000	2.3677121	0	12.0000000

Engineered features

The SURVEYSELECT Procedure

Selection Method	Simple Random Sampling
Strata Variable	default_flag

Input Data Set	PD_SORTED
Random Number Seed	2026
Stratum Sampling Rate	0.7
Number of Strata	2
Total Sample Size	71
Output Data Set	PD_SPLIT

PD model - logistic regression (training sample)

The LOGISTIC Procedure

Model Information	
Data Set	WORK.PD_TRAIN
Response Variable	default_flag
Number of Response Levels	2
Model	binary logit
Optimization Technique	Fisher's scoring

Number of Observations Read	71
Number of Observations Used	71

Response Profile		
Ordered Value	default_flag	Total Frequency
1	0	55
2	1	16

Probability modeled is default_flag=1.

Stepwise Selection Procedure

Class Level Information			
Class	Value	Design Variables	
marital_status	Divorced	1	0
	Married	0	1
	Single	0	0

Step 0. Intercept entered:

Model Convergence Status
Convergence criterion (GCONV=1E-8) satisfied.

-2 Log L	=	75.771

Residual Chi-Square Test		
Chi-Square	DF	Pr > ChiSq
33.4712	9	0.0001

Step 1. Effect delinq_severity entered:

Model Convergence Status
Convergence criterion (GCONV=1E-8) satisfied.

Model Fit Statistics		
Criterion	Intercept Only	Intercept and Covariates
AIC	77.771	56.316
SC	80.034	60.842
-2 Log L	75.771	52.316

Testing Global Null Hypothesis: BETA=0			
Test	Chi-Square	DF	Pr > ChiSq
Likelihood Ratio	23.4546	1	<.0001
Score	24.6649	1	<.0001
Wald	13.7070	1	0.0002

Residual Chi-Square Test		
Chi-Square	DF	Pr > ChiSq
13.9918	8	0.0820

Note: No effects for the model in Step 1 are removed.

Step 2. Effect dti entered:

Model Convergence Status
Convergence criterion (GCONV=1E-8) satisfied.

Model Fit Statistics		
Criterion	Intercept Only	Intercept and Covariates
AIC	77.771	46.859
SC	80.034	53.647
-2 Log L	75.771	40.859

Testing Global Null Hypothesis: BETA=0			
Test	Chi-Square	DF	Pr > ChiSq
Likelihood Ratio	34.9124	2	<.0001
Score	30.7086	2	<.0001
Wald	14.9798	2	0.0006

Residual Chi-Square Test		
Chi-Square	DF	Pr > ChiSq
5.0715	7	0.6512

Note: No effects for the model in Step 2 are removed.

Step 3. Effect employment_years entered:

Model Convergence Status
Convergence criterion (GCONV=1E-8) satisfied.

Model Fit Statistics		
Criterion	Intercept Only	Intercept and Covariates
AIC	77.771	44.634
SC	80.034	53.684
-2 Log L	75.771	36.634

Testing Global Null Hypothesis: BETA=0			
Test	Chi-Square	DF	Pr > ChiSq
Likelihood Ratio	39.1375	3	<.0001
Score	31.0560	3	<.0001
Wald	13.8351	3	0.0031

Residual Chi-Square Test		
Chi-Square	DF	Pr > ChiSq
1.4974	6	0.9597

Note: No effects for the model in Step 3 are removed.

Note: No (additional) effects met the 0.3 significance level for entry into the model.

Summary of Stepwise Selection

Step	Effect Entered	Effect Removed	DF	Number In	Score Chi-Square	Wald Chi-Square	Pr > ChiSq
1	delinq_severity		1	1	24.6649		<.0001
2	dti		1	2	10.7518		0.0010
3	employment_years		1	3	3.3878		0.0657

Type 3 Analysis of Effects

Effect	DF	Wald Chi-Square	Pr > ChiSq
dti	1	8.3793	0.0038
delinq_severity	1	13.2533	0.0003
employment_years	1	3.2508	0.0714

Analysis of Maximum Likelihood Estimates

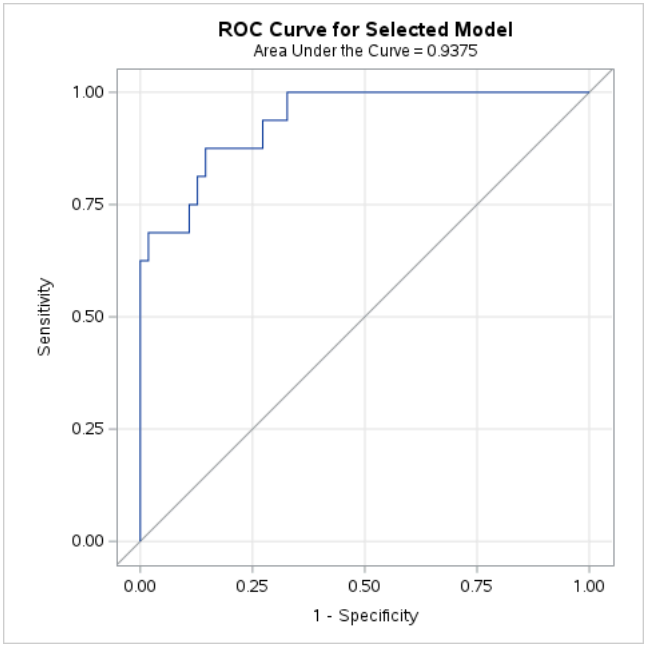
Parameter	DF	Estimate	Standard Error	Wald Chi-Square	Pr > ChiSq
Intercept	1	-6.5539	1.8692	12.2943	0.0005
dti	1	16.7051	5.7709	8.3793	0.0038
delinq_severity	1	0.9898	0.2719	13.2533	0.0003
employment_years	1	-0.2866	0.1590	3.2508	0.0714

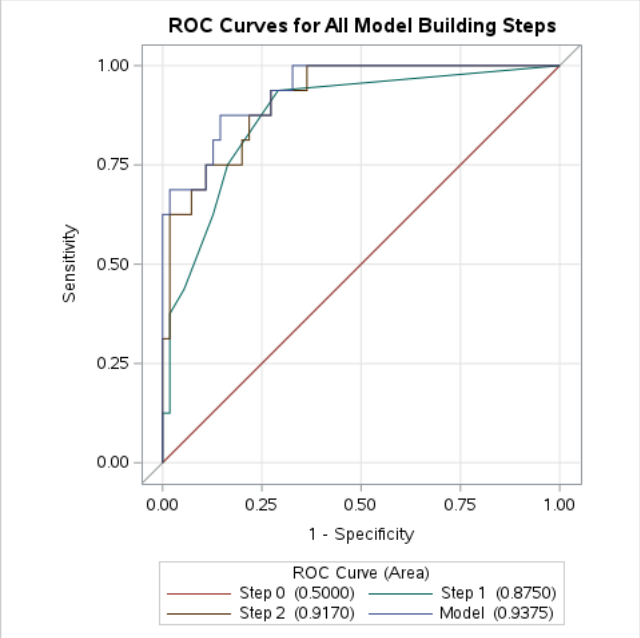
Odds Ratio Estimates

Effect	Point Estimate	95% Wald Confidence Limits	
dti	>999.999	220.149	>999.999
delinq_severity	2.691	1.579	4.584
employment_years	0.751	0.550	1.025

Association of Predicted Probabilities and Observed Responses

Percent Concordant	93.8	Somers' D	0.875
Percent Discordant	6.3	Gamma	0.875
Percent Tied	0.0	Tau-a	0.310
Pairs	880	c	0.938





Partition for the Hosmer and Lemeshow Test					
		default_flag = 1		default_flag = 0	
Group	Total	Observed	Expected	Observed	Expected
1	7	0	0.00	7	7.00
2	7	0	0.02	7	6.98
3	7	0	0.09	7	6.91
4	7	0	0.19	7	6.81
5	7	0	0.31	7	6.69
6	7	2	0.52	5	6.48
7	7	0	1.20	7	5.80
8	7	3	1.94	4	5.06
9	7	3	4.38	4	2.62
10	8	8	7.34	0	0.66

Hosmer and Lemeshow Goodness-of-Fit Test		
Chi-Square	DF	Pr > ChiSq
9.2814	8	0.3191

PD model performance (AUC / Gini / KS)

sample	auc	ks	gini
train	0.938	0.730	0.875
valid	0.761	0.609	0.522

PD calibration: predicted vs observed default rate (group 0 = riskiest)

risk_group	n_loans	avg_pd	observed_dr
0	20	85.60%	65.00%
1	20	27.35%	30.00%
2	20	6.84%	15.00%
3	20	1.64%	0.00%
4	20	0.16%	0.00%

Observed recovery rate and LGD

The MEANS Procedure

Variable	N	Mean	Minimum	Maximum
recovery_rate	100	0.3286585	0	0.9210993
lgd_observed	100	0.6713415	0.0789007	1.0000000

LGD model - OLS regression on recovery rate

The REG Procedure
Model: lgd_reg
Dependent Variable: recovery_rate

Number of Observations Read	100
Number of Observations Used	100

Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	6	7.64523	1.27420	113.41	<.0001

Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Error	93	1.04488	0.01124		
Corrected Total	99	8.69010			

Root MSE	0.10600	R-Square	0.8798
Dependent Mean	0.32866	Adj R-Sq	0.8720
Coeff Var	32.25121		

Parameter Estimates							
Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > |t|	Standardized Estimate	Variance Inflation
Intercept	1	0.20984	0.16937	1.24	0.2185	0	0
secured_flag	1	0.45593	0.06524	6.99	<.0001	0.73264	8.50178
coverage_ratio	1	0.08232	0.05107	1.61	0.1103	0.15191	6.86926
is_card	1	-0.05949	0.03689	-1.61	0.1102	-0.09248	2.54414
is_overdraft	1	-0.04142	0.03970	-1.04	0.2994	-0.05513	2.15868
is_revolving	1	0.00321	0.03851	0.08	0.9338	0.00353	1.39380
log_ead	1	-0.00314	0.01290	-0.24	0.8084	-0.00905	1.06983

Credit Conversion Factors by segment

product_type	util_band	n_obs	ccf_seg
Credit Card	HIGH	28	0.576
Credit Card	LOW	17	0.705
Overdraft	HIGH	12	0.615
Overdraft	LOW	18	0.788
Revolving Credit	HIGH	11	0.486
Revolving Credit	LOW	14	0.554

IFRS 9 ECL by stage

stage	n_loans	total_ead	total_ecl	ecl_coverage
1	62	16,366,031	722,134	4.41%
2	38	12,449,909	3,658,690	29.39%

IFRS 9 ECL by product

product_type	n_loans	total_ead	avg_pd	avg_lgd	total_ecl	ecl_coverage
Credit Card	38	5,796,485	13.23%	88.67%	627,420	10.82%
Overdraft	12	3,017,472	31.83%	87.03%	713,395	23.64%
Revolving Credit	16	5,452,283	14.52%	69.48%	668,804	12.27%
Term Loan	34	14,549,700	38.67%	49.97%	2,371,205	16.30%

Top 10 loans by ECL

loan_id	product_type	util_band	customer_id	outstanding_balance	current_dpd	eir	default_flag	undrawn_amount	pd_12m	ccf	ead	lgd	stage	pd_lifetime	pd_used	disc_facto
L0092	Term Loan	HIGH	C28647	1237000	0	0.1135	1	0	83.50%	0.00%	1,237,000	83.42%	2	100.0%	100.0%	0.5536
L0084	Term Loan	HIGH	C96157	748600	5	0.1264	1	0	95.35%	0.00%	748,600	83.26%	2	100.0%	100.0%	0.7426
L0073	Overdraft	HIGH	C31300	198400	0	0.1516	1	177900	96.62%	61.55%	307,897	87.12%	2	99.89%	99.89%	0.8683
L0005	Term Loan	HIGH	C32549	449900	15	0.1077	1	0	99.49%	0.00%	449,900	83.10%	2	100.0%	100.0%	0.5996
L0031	Term Loan	HIGH	C89742	1332000	0	0.1135	1	0	19.83%	0.00%	1,332,000	83.44%	1	89.04%	19.83%	0.9476
L0029	Credit Card	HIGH	C68835	231500	15	0.2920	1	179800	78.41%	57.62%	335,096	88.96%	2	78.41%	78.41%	0.8797
L0012	Revolving Credit	LOW	C12317	187800	74	0.2146	1	682800	26.34%	55.43%	566,276	82.85%	2	45.74%	45.74%	0.8233
L0014	Overdraft	HIGH	C64944	247900	5	0.2985	1	25000	99.98%	61.55%	263,287	87.07%	2	100.0%	100.0%	0.6758
L0032	Revolving Credit	HIGH	C72537	417100	45	0.2018	0	333000	19.52%	48.64%	579,066	82.86%	2	35.23%	35.23%	0.8320
L0019	Term Loan	HIGH	C14748	282000	45	0.0981	1	0	79.03%	0.00%	282,000	82.95%	2	100.0%	100.0%	0.5976